

PWD DMS - System Overview

Public Works Department Document Management System

PWD DMS is a web-based Document Management System for the Public Works Department. It stores official records (PDF files and images) across organizational Wings, with folders, retention categories, metadata fields, and intelligent search - so users can find documents not only by title or file number, but also by words written inside the actual document.

DEEP SEARCH - HOW PDF & IMAGE SEARCH WORKS

Normal search only looks at form fields (subject title, file number). Deep Search goes inside the uploaded file itself. When a PDF or image is uploaded, the system automatically reads the document in the background using AI-powered document intelligence and saves all readable text into the database. Later, when you type a keyword in the search box, the system searches three layers at once:

- Layer 1 - Record metadata: subject title and official file number (e.g. PWD/2025/...)
- Layer 2 - Deep content: full text read from inside the PDF or image file
- Layer 3 - Combined filters: category, classification, date range, folder, wing, etc.

Example: If a scanned register page contains a name or number that was never typed in the upload form, Deep Search can still find that document - because the AI already read and stored that text when the file was processed.

AI DOCUMENT READING - WHAT HAPPENS AFTER UPLOAD

Upload is instant for the user. AI processing runs silently in the background:

- Step 1: File is saved securely (PDF, JPG, PNG, or WEBP up to 50MB)
- Step 2: AI reads every page - printed text, scanned pages, and image photos
- Step 3: Extracted text (Urdu, English, Roman Urdu, numbers) is stored in the database
- Step 4: Document becomes fully searchable in Master Register within minutes

For PDFs: AI converts scanned/non-selectable PDFs into searchable documents and reads all page content. For images (phone photos, WhatsApp scans, handwritten registers): AI analyses the image visually and extracts visible text. Handwritten or rotated documents may take longer to process but are still indexed for search once complete.

USER ROLES & MAIN MODULES

- Admin: all Wings, dashboard, impersonate wing users, system-wide Deep Search
- Wing User: own wing documents, folders, categories, and search
- Dashboard | All Wings | Document List | Folder Manager | Category Index

HOW TO USE

1. Login at /login (default admin: info@pwd.com / admin123 after db:seed)
2. Admin creates Wings; each wing gets its own user account
3. Upload document with metadata (category, file no, subject, classification)
4. Wait 1-5 min for AI background processing to finish
5. Use search box - type any word from inside the document, not just the title
6. Combine search with filters (category, dates, folder) for precise results

Deep Search Tip: If a newly uploaded document does not appear in content search yet, AI processing may still be running. Refresh after a few minutes or use Re-extract to re-process the file manually.